

Lecture 8 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Part I: Individual Reflection

Work independently for the first few minutes. The goal is not to produce perfect answers. Be honest and specific about what you understand and where you are getting stuck.

1. Confidence Check

For each topic, circle your current level of confidence.

Topic	Not yet	A little	Mostly	Confident
Describing a line through the origin as the span of one vector	1	2	3	4
Writing a line in vector-parametric form	1	2	3	4
Writing scalar-parametric equations for a line	1	2	3	4
Describing a plane through the origin as the span of two vectors	1	2	3	4
Writing a plane in vector-parametric form	1	2	3	4
Writing scalar-parametric equations for a plane	1	2	3	4
Finding a vector perpendicular to two vectors in a plane	1	2	3	4
Writing an equation for a plane	1	2	3	4
Writing a line as the intersection of two planes	1	2	3	4
Working with affine lines and affine planes	1	2	3	4
Visualizing lines and planes in $\mathbb{R}^3$ using the Desmos 3D Calculator	1	2	3	4

2. What Makes Sense to You?

Choose one idea from the table that you understand reasonably well. Explain it in your own words, as if you were helping a classmate.

3. What Is Your Muddiest Point?

Write down the most specific question you currently have.

Avoid writing only “I do not understand planes.” Try to identify the exact step at which you become unsure.

Examples of specific questions:

- How do I find a vector perpendicular to both spanning vectors?
- Why does one equation describe a plane rather than a line?
- Why do we need two equations to describe a line in $\mathbb{R}^3$?
- How do I know whether a parameter should be called s or t ?
- What changes when the line or plane does not pass through the origin?
- How do I enter a parametric line or plane in the Desmos 3D Calculator?

My question:

Part II: What Do the Representations Mean?

Complete as much of the following table as you can without consulting your notes. Then consult your notes if you wish to fill out as much more as possible. It is fine to leave an entry incomplete if you are unsure.

Object	Parametric description	Description using equations	How would you visualize it?
Line ℓ through the origin in $\mathbb{R}^3$	$\text{span}(\vec{v}) = \{t\vec{v} : t \in \mathbb{R}\}$	$\vec{n}_1 \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0,$ $\vec{n}_2 \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$ where $\vec{n}_1, \vec{n}_2$ are in ...	
Affine line ℓ' in $\mathbb{R}^3$			
Plane P through the origin in $\mathbb{R}^3$			
Affine plane P' in $\mathbb{R}^3$			

Part III: Group Discussion

Form a group of four or five students. Each person should take about one minute to share:

1. one idea they understand;
2. one specific question or difficulty.

Listen for questions that several people in the group share.

4. Group Understanding

Write down two ideas that someone in your group explained clearly.

1.

2.

5. Group Questions

Agree on the three questions your group most wants the class to discuss. Put the most important question first.

1.

2.

3.

6. Choose a Spokesperson

Choose one person who will report your group's first question to the class.

Spokesperson: _____

Part IV: Class Discussion Notes

As groups report their questions, record the main topics raised by the class.

	Question or difficulty	Key idea from the discussion
1		
2		
3		
4		
5		

Part V: After the Review

Complete this section after the class discussion and review.

7. What Changed?

What do you understand now that you did not understand at the beginning of class?

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8. Remaining Question

What question, if any, do you still have?

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